

John H Clippinger & Andrew Pashea: "BioFirm Workshop part 2"

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all right welcome back we are here with John clippinger and we will be discussing active inference biofirm homeostatic agents for Biore Regional and regenerative Finance so thank you John looking forward to this presentation and discussion and more Daniel thank you so much for doing this and hosting this uh it's been a great set of sessions it really has been very informative and this is a followup on a session that we had yesterday on basically um what active inference agents and why they're important and uh and looking at them in an application and in this case in this session we were going to look at active inference agents uh in terms of a concept called biofirm uh as a homeostatic agent support Biore Regional regenerative Finance we're really looking at that the taking the thinking behind active inference agents and applying it to a new notion of the firm that's inherently biomedic and and what I wanted with these series of slides is to sort of set up the conversation and the discussion that we're going to have later with Andrew and Daniel on the providing an in-depth look at at biof firms as a homeostatic agent that is capable of interacting with the environment in a way that preserves it and pres preserves itself in the environment um so so what I'd like to begin with is sort of this a premise I've been involved in uh looking at regenerative finance and issues around climate change for at least the last 10 years and I think there's a growing consensus that we need a new kind of science-based Economy based an enhancing life in nature and quality you really need to think about what are the economic incentives uh to make a transition to an economy that values the preservation of life and nature rather than the extraction or destruction of it and what we don't need is what we've we're we're now feeling the the consequences of it and that's the commodification of nature in people turning them into objects and assets that can be bought and sold and trade without any discrimination between sort of what kind of asset we're talking about were you talking about something that is is coal as water as life the biosphere they're all assets uh they can all be priced and they all can be traded um and so you just have problem of total fungibility when actually some things are

essential for existence and they exist in relationship to other things to be preserved as living things um and the other problem we're seeing it experiencing it widespread just today sort of the unbounded sort of concentration of of of wealth and power so we have a system that works towards monopolies and the concentration of wealth and privilege and and this of course has created enormous pressures not only sort of on the the ecology but on the whole society itself and you have this notion of maximizing individual utility so everything is about Freedom the individuals in atomic agent independent of the context in which they live and their impact upon others not just other people but their other species the whole environment um and so we have extracted metrics that with negative externality so so know there's you can create quote economic returns but what you're doing is creating a negative externality would be pollution or it be uh air pollution destruction of species or habitat this is not reflected in the price and then it's left to some other party to a public good party to clean up the mess but this is they're not able to clean up the mess fast enough than it's being created so we have a sort of a zero sum win or take incentive structure and we have a lack of transparency and accountability so there's no sense of of collective purpose so th this is is really force people to rethink what an economy should be uh and and what it would be to have biomatic firms an economy something that that actually worked in concept with nature not against nature so first thing is important is not to have any negative externalities this idea that that you you can offset the cost of something is going to be picked up by other parties since their carbon credits and offsets are really an attempt a failed attempt to deal with negative externalities and we we need to get where we need to have nonzero sum incentives in other words incentivize the collaboration and cooperation of parties to take Joint outcomes rather than maximizing utility for a single individual and we need to have a new way of recognizing that value is created in in a really scientific incredible sense that Val is recognized when value is recognized through a bio capacity and diversity and resilience have that be a notion of value not just to something that maximizes a price Arbitrage but actually creates new kind of value the other important thing I think is is coming into focus is this idea of agency um and this is sort of A New Concept that thinks that ah nature should have agency we should have an agency and we should have agency of of of thinking of nature is being given the capacity to be itself and species to be itself and people to be themselves and not have that be restricted and with that I think we comes the notion of what we call sort of homeostatic fungibility in other words how something is is value of that realized is in relation to keeping other things Al life if you look at the the human body it's like you can't maximize the utility of the heart against the brain or the

lungs they all exist together so how you allocate resources in the body is really maintaining the balances between those different kinds of organs and that's true of Nature and that's true of organizations what we need also is a sort of scale-free and domainfree metrics that's scientific and adaptable so we really need to have the scientific process be a fundamental organizing principle by how we recognize and and and and and exchange value um and this means having something that's really is transparent explainable you can't have something that's a black box that's not opaque that you don't know where it comes from you don't know how to trust it you don't know how to derive it it really needs to be something that's transparent and so we've seen people talk about the blockchain and things like that but it's more than just a blockchain it's a whole set of methods being able to see how something is derived and be able to replicate that and then the idea of scalable Learning Systems should learn and they should improve over time markets don't necessarily do that they just they actually they they they they have clear a market around a price and then they start again there's no memory there's no learning how to do something better for adaptation as as a people and in the planet we need to learn we need to learn improve upon ourselves and that should be part of our whole economy not something that's tangential to it and finally the whole idea of governance there's governance should not be something apply after the fact or top down it should really be embedded in the whole system of itself and so this's been a lot of talk about decentralized governance and decentralized system decentralized autonomous organizations but a lot of that is just talk they're not really fully autonomous and not fully decentralized but how do you really realize that that vision and these these ideas have really sort of were first I think articulated and recognized um by The Economist uh elanar a who got a Nobel Prize for her work um on the governance of econom and she was the first woman Economist to do that and and I'd worked with elener when I was at the Harvard the burkman center and looking and and trying to get thinking about a digital common sent it was too early but here are some of the central principles that she had and this is back then when the technology was not very developed but she developed her thinking based upon how indigenous people managed resources their common resources and that would be their their their their forestry the Fisheries the things that they've done for thousands of years and they're able to keep them in balance and keep their lifestyle in Balance through that and here were some principles she extracted from that so was clearly defined boundaries um and you needed to have a a common po resource where people recognize as a common po resources you have to have it congruent with local conditions and I think be that in other words you have to recognize each little bio

region had its own PE uh constraints and and and issues that you needed to recognize and you had to have ways of people could make Collective choices among up Commons how bring people together they couldn't be anonymous one they really had to be they had to know each other and they be accountable to one another and you had monitoring process you had to monitor it and be able to see where you were and then if something goes wrong you have what she called graduated sanctions which means it was you you the punish was was commensurate with the degree of the violation but you really didn't want to have a highly punitive system that its own negative results and you had various mechanisms for re resolving conflict um and then I think the the the idea one of the things she her concerns with particularly with indigenous people is that the legitimacy of those organizations were recognized by the larger nation state that that that they had some genuine sovereignty and integrity and finally this whole idea of nested Enterprises so recognizing it wasn't a strict hierarchy there was things that dynamically interacted at different scales and these were ideas that were sort of more metaphors at the time they were aspirational metaphors was very hard to realize in any kind of operational form and she went on to talk about later work in social ecological systems and recognized the biophysical and the social factors how they all work together and V and how you look across different spatial temporal scales and different hierarchies how they're linked um and she was always starting to think about what were the dynamic complex systems that can characterize by a adaptation if you're going to have governance but where there really wasn't a mechanism there for for how to realize those ideas now there's another more recent development uh is actually it's called tanu that was started by a friend of mine um name is jonath lagard is in Africa and they're doing this in Rwanda and they're doing it with all kinds of species are they starting out with the gorillas or W but the idea is is that animals species contribute to the environment and they should be paid for the value that they created so how do you do that you're giving agency not only to the animal but to the to the habitat in this case in Rwanda they want to give not only agency to say gorillas Rwanda gorillas but they also they want to give agency to a rivers in in certain bio regions and by agency it mean the the fact that they these things is staying alive can be valued for their capacity to stay alive and create a habitat that creates a a successful and and resilient environment um and so what is this is a quote from uh their their literature their website which I recommend you taking a look at and exists with the belie that for these species leave they must actively participate in the economy so how do you create a new economy that includes living things and and and how do you how do you recognize that value another approach is something that's come from regen networks which

has been working on this for a number of years and I've worked with them and they're talking about creating ecological institutions and again it's like what is an institution an institution in the classic sense is separation of people from a set a set of principles like the rule of law you say well here's an institution and it's set up to do these things and people to perform certain roles in an institution um but the problem with institutions is that they're run by people people get corrupted and they institutions don't work in this case what they're trying to do is develop a constitution for they call an ecological subject and again is giving agency and recognition to Nature um and then they have a number of of methods by which to to govern that and put in a ledger create some token system Region's been doing this kind of work for quite a while all part of a common effort another effort is is something called a bi Regional Hub and this will be my last example before making the transition um but the the the idea of a b Regional Hub is that to pull together different parties in a particular geospatial location and be able to give them a a set of finance and incentives by which they can act to actually create the bio capacity of that particular GE region and so it's really working with core communities that that are committed to actually en enhancing a could be a river a river system it could be the one of there they're looking at in Northeast uh bio region is the Connecticut River how can you bring it back to where it was with abundance when they had full salmon they had was a very rich ecological system but there are other things in terms of land owners and being able to give them the incentives to rewi their property and actually make a living on that and create an economy an exchange of value in such a way that makes it robust and resilient um and so they're working with different kinds of investors and social impact investors in order to make this kind of happen so this is part of a larger collection of people are trying to do these things now what we're going to say what I and this is a very cred I apologize for the the verbiage here but the we're and and what you're going to hear in this the subsequent discussion is this idea of a biofirm is a bio which is is a firm that is based upon homeostatic principles and it really I think addresses the many issues both operational and aspirational that you see in the whole bof Finance movement and and and to to to run through this fairly quickly but you're going to find Andrew and Daniel going to talk about these things in detail with examples is that it's first principles based it's grinded in scientific method basing in statistics and markup blankets in other words you drill down to it and you really understand it rigorously as a science you can understand where what it's based upon and why how things work in a principled way it's also computationally tractable at scales and and with free metrics for doing uh the whole kind of reporting uh and and uh that you you have this required environmental reporting they call

mrvs um sort of uh and so do you have a method by which you can make statements and claims about the state of the environment that is scales that scales and and and can be used for cross comparisons it's also going to be its open source so the bar form is like a kernel think of it as a Linux kernel for a new kind of organizational structure can then generate a new kind of economy that's Biomet biometrically based another key component is if we're going to scale our capacity to deal with nature it's going to be a lot is going to have to be automated it's there's so many tasks involved in different parts of the world that you have different skill sets how do you how do you automate some very complex things in a in a way that can be trusted and generate metrics that other people are willing to accept this has been a big problem in climate change and this is where the the an extraordinary automation of cognitive tasks through through large language models Transformers and agent architecture allows for an automation of many many of the tasks of collecting and analyzing the data and learning from the data this is very significant um and then you can create you have the capacity to have create networks of Agents working together and create a system exchange so that's a Natural Evolution that's happening into edge-based Computing that you can have Computing on the edge you can security The Edge that's another phenomena and then I think you know the the idea of polycentric government this not is made of that but what does that really mean well really means if you look at agents that are self-organizing agents and dynamically relating to one another on the basis of the free energy principle these other principles you have the effect of of of a polycentric governance um and you also have the ability to formally give agency to Nature to different species to different peoples to different kind of competencies so that is the whole nature as to say of of an agentic architecture um and then I think you you have a principal means of interfacing between a biometric economy and the the the current economy that's going to be really important so you can't just live in your own little world you're going to have to interchange with the the Fiat economy of the world in such a way that actually protects and and establishes the value of a biometric economy finally I think it provides a a a principal basis for valuing nature-based currencies and assets that we we can discu a little bit in the time to come so that that is that's basically uh my my comments I think that I like to end is said let's make it real that we're actually I think you'll see in the following discussions we're at a point where I think the technology is getting sufficiently mature to actually engage in developing use cases of this and bring it to scale so with that I I'll hand it over to uh Andrew thank you great thanks John um quick mention to Daniel uh John and I had a a really nice conversation prior to this if it's possible uh I might actually start my talk now uh and then yeah perfect I was

hoping that would be the case and then that way John and I can also have a little bit of back and forth potentially during the course uh and obviously Daniel you've had so much a big hand in the development so anytime you know yeah would like to chime in please feel free great um all right so my original plan talk for today was entitled uh let me get rid of this guy uh active inference is a framework for social science-based uh modeling and okay um something happened on the way to the Symposium yeah exactly uh huge updates uh I I want to just give a little bit of context uh for some time now I've uh you know I personally gave a talk a couple of months ago a three point a three and a half hour Workshop basically at a International Conference for computational social sciences uh hold at held at University of Pennsylvania uh July 17th this year and uh what what I was trying to do was basically teach uh computational social scientists that means economists sociologists political scientists and so on who are very Savvy with computers essentially um how we might be able to employ active inference uh in agent-based modeling which is already you know a decades old at this point uh method of Designing uh uh basically simula of Agents inside of environments where agents might be you know human beings and an environment might be you know a market system or or some other kind of environment that you know uh humans find themselves in where you can you know you can play out simulations test things and so on I want to introduce that to computational social scientists and I think more broadly what's what's great is that this Symposium now gives me the opportunity to kind of flip it to where I can kind of bring some of what computation social science does to other active inference researchers um but to stay on with the biofirm uh I really wanted to again as Daniel said something happened on the way to the Symposium uh I've had these conversations with with John since uh since I did that conference and uh I have a lot of kind of strong intuition that a lot of what he's been describing is is a really wonderful sort of direction for future work in both agent based modeling but also applications of that kind of in the real world um I mean we're always operating under conditions of you know uncertainty and and trying to figure out kind of what to do in very adaptive ways where our environment is is constantly changing uh from you know implementation of computers social networks uh we have climate change we have uh the typical booms and busts of our economic systems and so on so um so I'll I'll start here but I just want it to be known like there have been a lot of heavy revisions to this with a lot of development we've made on the biofirm D featuring biofirm uh so again I'm Andrew pase uh I my day job I work as a data analyst scientist engineer always many roles to fill uh and so uh I I bring a lot of those kind of skills with me to active inference and how I view things like developing um kind of applications uh

for specific use cases and things like that I think there can be a lot of Rich uh interdisciplinary dialogue between you know different fields and that includes the kinds of things I do on a day-to-day basis versus uh my interest in active inference which has been going for the past few years now so I've I've been a an intern and I I suppose I'm feeling much more like a like a full-time uh kind of affiliate of the institute at this point I've been teaching the the text group um uh with Daniel co- facilitating that I've been involved in various projects uh and so and then now this kind of partnership with with first principles first and what uh Dr Cliff and Jur is trying to accomplish and and developing the biofirm so I just want to put uh you know on this title slide here these are actually links to you know if you would like to get involved with the active inference Institute with uh check out all the kinds of things that are being put out by first principles first and also I just want it to be known like you can come back to these slides as needed as well as any of the other kind of learning materials I've been produced saying whether it be for the the conference that I did a couple months ago including on the code uh how to actually create active inference uh Agents from scratch uh moving up to multi-agent simulations and so on I also did a talk with the Institute recently that kind of walks through some of the some of the code so um the the new outline for this talk we're going to touch again on agent based modeling which we already talked about yesterday uh a little bit in h the first Workshop so I'm we're going to kind of skim through that a little bit but I W to I want to put some special attention upon the aspect that uh of Agent based modeling referring to what are called micro assumptions about Behavior Point Blank if we're going to simulate humans we have to include certain assumptions uh about how we think humans behave if we're going to simulate them in an environment right otherwise we just we we can create a list of humans they do nothing until we give them some kind of rules or behavioral you know Dynamics or something along those lines so those are essential uh for for setting up agent based modeling and then uh and then we'll shift over into active inference we'll look at the PDP as a kind of Exemplar model talk about some agent multi-agent examples just to give a sense of what kind of work be done in this area many people watching this now who who are already involved with active inference there'll be some aspects of this that are you know might come off as a little repetitive but I just I don't want to I I I want to stick to the the textbooks so to speak right I want to kind of keep this grounded in a sense that you know folks who know about active inference don't know about active inference have some vague interest in active inference in this in between area like can catch on to the material here um next we'll go into this question are llms agents uh there's a lot of exciting work being done with llms uh they're showing

incredibly uh you know exciting and surprising uh you know sophisticated reasoning uh skills and capacities for dealing with natural language that said um you know and it's already been said so many times in the other talks given in you know during the Symposium but it's worth repeating like they have limits there are issues and I and I've been doing some work on thinking through how do llms relate to things like cognitive theories what how does an llm think or how how does that work really so that we can really grasp if we want to incorporate llms into say agent based modeling software applications or otherwise let's let's come to terms with like the reality of what they are before assuming too much about what they are or what they're capable of and furthermore how we might to integrate them into kind of more grounded means doing agent-based modeling such is working with actual agents the quick answer to our llms agents agents are they agents no that that's the quick answer they they are not uh that's what I will claim and then we'll move on to the biofirm which frankly will take up the majority of the rest of the talk uh from there so uh what is agent-based modeling uh as already stated we have agents more specifically uh you know it's a programmed entity it could be a person a living organism it could be representative of an entire institution uh or otherwise uh and basically programmatically it receives observations from its environment and then it returns actions back into its environment for example an agent could be a person an investor and then uh with the environment uh the the environment could be an artificial or real uh system that elicits observations agents and then receives the agents actions uh those actions then impact the environment itself in turn for example the environment could be a stock market so we see that there's this reciprocal relationship between the two things an agent or environment's action is an environment or agents observation so the agent sees a price signal from the environment they make a purchase uh next morning prices have adjusted given that agent maybe other agents purchases you see so there's this kind of back and forth in a loop uh you know a certain time interval such as in time steps or we could imagine them in days like on day one this happened day two this happened an active inference the agent environment's respective Markov blankets which kind of Define what it is to be that entity whether it be the agent or what it means to be that environment including all the the the structure uh the characteristics you know if it's a person that maybe it includes things like their goals or or their um uh what it means to maintain homeostasis so an individual has a you know blood and oxygen flows in their body you know what does it mean to be that entity so from there we simulate them we play out the agent environment Dynamics over time again as I said in time steps some other kind of interval and uh and in active inference we often use the phrase action

perception Loop you know it's like a you do something then you perceive the outcome you elicit your next action you know receive the outcome and then you know arguably this happens at a very rapid Pace in actual human beings right this is goes beyond just time step one or two or looking you we measure this in in in Nan seconds or otherwise so core guidelines setting up agent based models we want to make realistic yet simplest possible agents and environments that has to do with you know uh generalizability if you make something that's so unique to a particular situation it might not generalize well uh you know so if you want to make an agent agent based model of say uh you know agents taking care of a natural environment if you make the agents way too specific to that environment to where they you know if you if you want to set up this agent-based model get insights into what you can do in reality you want to make sure that what you test in the model can actually be carried out in reality can't be too specific it can't be off course um robustness simply small changes don't dramatically change results uh so you know if you fine-tune something too much such that you know the the results of the um experiment end up being dramatically different and suddenly once again we have that generalizability problem reproducibility everyone knows reproducibility problems exist in modern science we need to document and publish uh you know our code make it available to other people so they can understand it recreate the results Etc and finally usefulness which is simply why do this you know if we can't if we can't get any kind of additional useful insights out of setting up these rather complex computer models and and programs um you know why do it and so it's it's worth having that in mind you know not not all models are right uh but but some models can be useful right in the statis and George Box's words then um purpose so we do agent based modeling in order to predict outcomes we could try and extrapolate from the present uh we can entertain different kinds of hypothetical scenarios design new policies and and have them play out over the course of the simulation to see what the outcome is and then we can also look at Collective outcomes of aggregated individual behaviors if all uh you know stock investors in my simulation All Began investing in this particular stock like what happens by the next day do we end up with some kind of bubble and then bust kind of situation like what what occurs you know that kind of thinking and then over the past couple decades and this is really what I want to highlight a couple months ago and what I would like to highlight now again and I think it's especially relevant for anyone wanting to do agent based modeling and especially folks who are interested in active inference there's been a cognitive turn uh you know a lot of these traditional agent based models you know they're very you know very interesting a lot of thought has has has been put

into them but their agents tend to be very simplistic um in part because you know 40 years ago we were working with uh you know much more at this point archaic uh computers and computer architectures and and devices that simply could not run uh simulations that involve many interacting agents who all have highly sophisticated you know Behavior furthermore it's called a cognitive turn not a complexity turn but a cognitive turn in the sense that we're looking at actually more closely approximating human behavior so that means actually looking at other fields ranging from you know Neuroscience cognition psychology social sciences I means looking at uh you know cybernetics uh biology you know trying to open up the the discourse and the dialogues over how can we more better approximate human behavior so there's some common characteristics for these calls to cognitive agents that includes like having inter you know agents have beliefs um they update their beliefs based upon what they see they have to infer what to do in a sense that they're more autonomous they they kind of make their own decisions traditional agent-based modeling it's like well if if if if the agent sees X then they do y otherwise they do Z and these highly simplistic you know programmatic rules that you know the program you know I program them that way so I already know that they're going to do that uh whenever it comes to agent based modeling with cognitive agents I don't necessarily know exactly what they're going to do they kind of make their own decisions right they have the architecture for that and then also reinforcement learning which is kind of a you know adjacent or subfield uh in some ways of like machine learning data science you know a lot of these kind of state-of-the-art but still rather standardized at this point ways of doing modern data uh you know processing and analytics and and predictions and forecasting uh reinforcement learning is the way of doing that where you actually have an agent all right so um so that's been kind of one of the main ways people are trying to to continue this cognitive turn and agent based modeling now micro assumptions there a lot of information for a single slide so you know it'll be over soon but I I really just wanted to give uh um you know kind of a picture of there there is so much research that's been done so many more kind of dialogues between fields that that can occur uh here now um like when thinking about how to make cognitive agents what makes sense you know what what what's realistic but what also balances complexity and accuracy keeps things simple enough to where the agent-based model can actually be useful so there's been a lot of criticism at this point centuries criticism of this kind of neoclassical tradition classical than neoclassical tradition Adam smithian kind of tradition uh that tends to paint humans as utility maximizers you know it's often criticized because it's too simplistic it reduces people to being uh greedy or immoral you know as if

they're just going after you know profit without any concern for uh any other you know condition or needs or wants or you know um so kind of selfishness right without any form of Altruism or S of community um some people point to Economist Lionel Robbins you know kind of a classic in the field uh who claim that unlimited wants scarce resources that is the fundamental economic problem yet this assumption I think this gets overlooked I also criticize the utility maximizer assumption but I I want people to be aware that this this leads to a really useful tool which is for agent based modeling you can severely reduce computational costs for putting together agent based models if you just have a bunch of agents who are trying to act to maximize a singular value there's not a lot of complexity there you can you can run those equations quite quickly right and then that also assumes that the main thing in your environment is the singular value that is utility or a stand in might be uh you know money or or some general notion of liquidity and so since then we've had so many other ways of looking at Human Behavior Uh behavioral economics and finance psychology cognitive science social theory you know humans aren't just trying to maximize utility they're they're they're speculative they don't have fixed preferences they actually look at one another's decision makings you know this goes back to you know John Maynard Keynes we had himman p Minsky who looked at uh you know Market crashes we have Robert J Schiller who's still alive today in writing on narrative economics the idea that you know ideas have an impact on how we interact with our economy and how how we behave um there's conans thinking fast and slow like sometimes we just rely upon heuristics or maybe we do it a lot of the time you know it's quicker it's it's just um it mitigates having to put a lot of thought into things where we just kind of want to get something done even though it's not necessarily maximizing utility um there's star and Stein's nudge it's like sometimes yeah you do just go with the default option some people whenever a website asks them like do you accept all the cookies they will click yes quickly just because they want the thing off their screen right um so and then we have social emulation Amis convergence ideology all these theories um you know what is what is money it brings up the question of what is utility what is money currency actually isn't just a means of exchange it's not just you can also a store of value as its own value changes to to you know TR try and do kind of economic transfers to make sure that you get higher value out of whatever form of currency you're holding uh it's also a social institution though it's much broader right so be having access to money as many of us know allows you to be uh first you have access to a variety of Commodities assuming you're in a market economy where many things are fungible and tradable in that way and then we also have social belonging it's very

difficult to maintain yourself in a society where you have no money right so there it's a whole dimension that that is lost we just look at something like maximizing utility uh you know I've put a mention of Lou alus which is a much more kind of critical theory approach but it looks that the impact of state law uh you know different forms of coercion upon people to to act and behave in particular ways that go beyond just their ability to to try and maximize their own profit the pragmatism people up ate their beliefs they're forward-looking they don't maintain the same preferences and ideas and rules uh you know throughout history you know they change they have their own circumstances and so there's a kind of pragmatic uh approach to life Carl pan economic activity is embedded in society uh meaning that a lot of the AC economic activities that we have you know they themselves are predicated upon our ability to trust one another work with one another and all these other kind of things where whenever that trust is harmed it can dramatically change how that economic activity plays out and then we have Neuroscience biology physics and that you can see how we're moving more in the active inference Direction now but you know that we have these are predominant theories now in neural science the idea of predictive processing that you know that one comes to a situation with their prior beliefs and they they they can be updated in the sense of the the Basi and brain offis a kind of way of mathematizing that that process we also have and this is quite old now but it's still still stands as a role like heian learning or associative learning you know so someone who Associates money with being something bad very well might not you know have such a drive for money versus relative to other things um and then homeostasis allostasis allostasis being planning for homeostasis or homeostasis in the future these are obvious things for us that we deal with on the day-to-day we need to to fulfill our needs uh physiological needs neurobiological needs uh unconscious processes in the brain you know that goes beyond being able to say that every single thing that we do is necessarily conscious or deliberate right I I'm not actively trying to think about how to regulate my blood or oxygen flow right now uh we also take preventative actions you know we do things that might be a cost right now but they change you know the impact of whatever happens in the future for us uh we are forward looking in in various ways and we do often rely upon habits as needed uh to fulfill our needs habits allow us to do things without having to put in too much expenditure mental expenditure you know metabolic expenditure to ensure that our needs are going to continuously be fulfilled and cybernetics embedded cognition uh you know PE this is the idea that behavior and even intelligence arise from interactions between brain body and environment which kind of overcomes some of these uh ideas you classical ideas of a cognition mainly being uh outside

in where you just kind of look at what's around you and then you take that in as a passive Observer um instead we're actually always in communication with what's around us and Mortal computation in the sense of we can move in the direction of you know if we want to do something like cognitive modeling then we're looking more at uh you know actually trying to mirror these kinds of processes they're much more realistic collaboration imitation computational Neuroscience Psychiatry and modeling modeling internal Dynamics in form of you know time series neuroimaging data uh you know applying different kinds of techniques like causal inference to see how one factor impacts another one variable impacts another um actually trying to find correlations between internal neuronal Dynamics with actual behaviors that occur uh you know so so for example sticking you know kind of electrophysiological like electrodes on you know a mouse and a in a tze to see like okay what kinds of neuronal Dynamics correlate with you know the rat choosing to go left to find the reward uh you know find the cheese or otherwise and then and then taking that and trying to move in the direction of you know creating an agent who who basically is fit to the behavior of that that Miler rat in the team ma right that's a classic kind of example that's in active inference and we also have the hamiltonian principle of least action much more in the realm of physics um and it's it deserves a whole you know slide or 10 you know on its own but the the main idea figuratively is you know the idea of uh something occurring uh based on following whatever the easiest path is from one state to to another so a lot of these things the idea of you know social homeostasis uh you know relying upon horis sixs versus planning and forward-looking versus dealing with whatever's going on in the moment just because you need to figure out what to do now even though it's not something you plan for all these kind of aspects start to come together I argue an active inference uh so it's kind of like you know is this should we be considering something like comomo economic is revised you know again liquidity is the means for all Commodities at least in our market economy at least assum assuming fungibility of things that liquidity can get you many many things in a certain way I almost want to dial it back on saying it's immoral to want liquidity uh it allows you to fulfill your needs again in a market economy by seeking out Necessities for your own homeostasis or allostasis you know physiological relief or otherwise even then even as your bank account increases that doesn't necessarily mean you're going to have your social needs met kinship or familial needs met physical health uh you know if if a doctor tells me after I've suffered some kind of injury that I actually need to do stretches now in order to make sure that my arm will move going forward in the way that I need it to it doesn't matter how large my bank account is I'm still going to

have to do those stretches right um so there this a simplistic example but all of this is to say uh you know for agent based modeling rather than thinking of utility especially utility in some vague abstract sense of you know is it money is it liquidity is it something else there could be a shift to a kind of alternative uh you know what is essentially a virtually textually Universal value computed by the brain which can Encompass not just to drive to liquidity but other aspects and means of Life simple as possible yet no simpler and then it itself becomes a simplified recordable rip metric for measuring one's capacity for current and future survival uh and so that you know enter the energy principle which is one of the central principles of active inference uh here we're minimizing quantifiable uncertainty uh you know that by minimizing this it allows us to achieve our goals fulfill our needs we learn we actually change our minds you know we learn new information we are not fixed beings who you know are are are kind of one-dimensional we update our beliefs and uh free energy is a proxy for uh uncertainty or surprise you know it's it's computed quite easily uh you know because it's a it's a variational quantity there's a approximate computation which you know in in physical living beings that keeps our metabolic costs low uh and and for agent-based modeling that also means keeping computational costs low and so it still kind of fits you know that the the the usefulness of what utility was for agent based modeling I think that free energy can instead be the new kind of metric that we're looking at um and you know I included this kind of a you know fun fact that there was a nice talk given uh not too long ago with active inference uh Institute uh you know some folks who do neuromorphic Computing just reminded us like despite how many calories their bodies use on the day today the brain only requires roughly four bananas worth of calories despite the fact that it makes 10 to the 15 uh calculations per second it's just a reminder of like this this variational aspect being able to do quicker approximations and then um you know we talk a lot about free energy but I just I really you know in the in the kind of funniest kind of way I just uh have always been fascinated by this equation uh this is you know textbook equation uh from active inference but you know so G is our expected free energy related to deliberative planning goal attainment information seeking conditioned on our homeostatic preferences right so we plan for things to care of ourselves to to accomplish things we need to F is variational free energy another quantity Where We balance the complexity and accuracy of how do we deal with the moment then we can think of this as like you know say you're camping in the woods and suddenly there's a bear it's like well you didn't plan for that uh and you didn't expect that and coming up against a potentially dangerous bear is probably very much against your preferences for maintaining your homeostasis in this case

your sense of physical safety so you have to deal with that and then e uh you know those are your habits your horis so so what I'm saying is that I just I'm particularly appreciate that we we have a quantifiable way you know here of trying to decide what uh you know what it is that determines someone's actual Behavior you know that the posterior beliefs High meaning what policy or action should I take right uh all of that is kind of all all these different considerations habits deliberative planning momentto moment planning all these things are are involved in that so it's just it's kind of it's it's an attempt at from my perspective like we're actually combining many of these inter disciplinary perspectives into a single equation there and so active inference I think I I I appreciate this attempt to try and do this kind of mapping of you know sub you know neurobiological substrates to actually interpretable cognitive theory regarding what is a habit how does that you know what what composes that in the brain what are goals what is planning what is action and then from there moving on to uh you know these kind of Exemplar uh initial active inference uh models which themselves are highly relatable to to reinforcement learning so it's all to make the case that active inference I think it draws connections and kind of builds Bridges between all of these uh you know these other disciplines in a way that that is very rich and it's not so uh exclusive right it doesn't it doesn't stop conversations it starts them so and then I also wanted to show like here is just a quick list of some multi-agent simulations with active inference that I think you know heavily relates it is agent-based modeling you can open the a textbook from a couple years ago agent-based modeling is there you know it's this isn't foreign to to active inference but I just I hope those in the agian based modeling Community can start to see how strong these relationships are with active inference so um epistemic communities under active inference this this uh paper was um you know recreating agents who hold their own opinions uh about things and they're they're in social networks so you can study how they share beliefs with one another and yet given you know a lot of these kind of formal principles and active inference seeing how that plays out we can actually see what the kinds of dynamics that we start to see on Twitter and other social networks where you know opinion polarization begins to happen depending on uh how you know how the network is set up in the first place so um you know it's just it's very interesting we also have active INF for ants this is much much more kind of a biological or Zoological example uh you know where there're ants in a colony who who uh have to adjust to changing context such as uh changing food locations and they leave one another pheromones so they can um actually follow one another's Trails so to speak to find the food and come back um and then this is very recent but uh we had another talk earlier today but the um this paper

with Connor Hines who is one of the developers of pmdp which we'll be seeing shortly uh he and you know Frisén and others are developing further tools for doing active inference and this kind of gets at the question of like nestedness and looking at how um you know you have a bunch of individual agents what where are the cases where their Collective Behavior actually you could fit that behavior back to the individual parameters that each of the individual agents hold um and then again with social science like looking at agents who have distinct roles uh such as leader and follower where or it could be you know you could apply this logic to many other situations the ideas just looking at how agents to have their own distinct uh you know uh available actions and beliefs nonetheless uh you know whenever they're kind they can kind of synchronize in their capacity to fulfill uh you know a goal that they collectively have um you know so it kind of speaks to the idea of having heterogeneous agents with heterogeneous you know concerns but nonetheless with an umbrella Collective concern they're able to accomplish their goal um this next one uh rather long title active inference framework for Collective parallel problem solving on Inky I States this is a you know I gave this at the tutorial a couple months ago um I just recreated uh so I'm this was the point I was trying to make you can recreate a lot of these old traditional agent based models you know using active inference agents that's what I'm claiming so so I'm not trying to say oh all of these traditional models are obsolete uh not at all I think there's so many fascinating design principles I would love one day to recreate the uh the Santa Fe Institute's artificial stock market but with active inference agents right we can do that so um so here I just recreated one where you have agents who are collectively trying to solve a problem represented on an Inky landscape meaning that they're trying to solve a very complex problem where there are a lot of interdependencies in different subparts of the problem you know it's like uh trying to uh trying to design a car it's like well you can have a great car but if you remove the wheels it's no longer going to really be the kind of car that you need right or you can remove the steering wheel right you need to have a certain combination of pieces there to have a proper solution so agents collectively do that and you can study like well how well do they do that do they get stuck you know with a suboptimal solution uh and then furthermore you can look at metrics like oh if we run this in different ways uh what is the connectedness needed uh to for them to reach the optimal uh uh solution or otherwise so and then finally this is not an active inference uh paper um but this is this is just a reinforcement learning uh you know based problem but I included it here because I just want to make the point that this is this is focused on industry right this is focused on multi-agent settings where you can have them kind of work

in M manufacturing firm that involves like making a lot of kind of Rapid decisions in order to uh manufacture particular products how you could have agents kind of synchronized in a way that they can accomplish you know that goal and uh again similarly with with traditional agent based modeling where you could you know swap them out with active inference agents so to speak might be able to do this you know take that same approach with looking at how people are using reinforcement learning today I think it lead to very interesting results so I just again just trying to bridge these kind of uh you know dialogues between people so now we'll move to llms as agents um this I just think it really needs to be uh addressed um so llms have been uh increasingly integrated into a wide array of potential use cases uh you know I I included some resources here like there there just uh you know dozens on dozens of of research papers that are being published uh that try to use llms in these kind of agentic ways where it's as if an llm is an agent who you know supposedly makes decisions autonomously and so on um again they're showing more and more sophisticated reasoning capacities um they're they're they're having uh kind of longer uh context windows and output Windows meaning you can feed them even more information at one time they can release even more or output even more information at one time uh you can also upload your own uh text files documents write your own prompts for them to respond to so in that qu in that case if an llm uh really is an agent then why not just skip all of this stuff with reinforcement learning with active inference why not go straight to using LMS as agents of course I'm gonna argue like we should not do that that be um I mean it would it would be irresponsible at this current moment in time in terms of llm development I I would I would argue that that would be kind of ir responsible unless you're just kind of making you know some kind of you know entertaining or or fun app that uses llms or something but for for something that that that calls for you know much more discretion I would not do that so what what are the biases here um uh llms are trained on large corpuses of existing text we all know this uh what that does is it kind of sediments a series of cultural scientific explanatory rhetorical Prejudice information figuratively the snapshot of the retrievable Web scrapable Internet other digitized human and AI produced text to which uh the llm's internal parameters which are you know blackbox parameters now we have llms being trained to where they have seven plus billion parameters on the inside um you know and then consequently the their outputs are just kind of reflective of this so so a lot of these uh you know prej prejudices have been addressed you know very obvious in a certain way very unfortunate things that we saw with initial LMS were like racist you know or or sexist or otherwise kind of outputs coming from them based upon the materials they were trained on

people are trying to find different ways of training them differently to get them to elicit the responses they uh that we want but that's the thing what's crucial when working with LMS is the prompt cell so I you know I I want to think you know just put yourself in the chair you know facing a computer and you're using an llm what's really going on you as the user you prompt the llm after it's already been trained you have your own purpose or goal in mind that the llm does not hold for itself quick mention I'm aware that I'm doing some slideshow karaoke I just for anyone who's like H but uh to know there's a lot of information I want to make sure I cover it thoroughly uh so so the user must provide the context for this goal uh or purpose and their prompts in the hope the llm will catch on less figuratively they hope that the llm will be able to predict the series of words and sentences which correspond in the user's final opinion to the answer the user needs or wants right and so after training the LM it can't actually learn what the user needs it's it's always in the user's opinion and we'll get more into that again it's just making the point The Prompt itself is incredibly important this is kind of a simplified uh ever so slightly exaggerated uh you know illustration of the kinds of things I'm talking about so if I go to an llm and say hi my name is Andrew it says hi Andrew it now knows my name how can I help you today uh I I ask it help me prepare you know I have a certain goal in mind I tell it help me prepare an initial draft for a proof of business model you know for an idea I have uh you know here are the following details that should go into that that business model uh model then it produces it for me and today it can even release it as a text file and so I could click on it download it great now what happens whenever our conversation history is deleted what happens whenever we exceed the context window that is repeatedly being sent back to the llm every time I prompt it um we have context loss so whenever that happens the llm you know I say okay now please help me revise the file please provide the file you mentioned it does not recall anything that came prior right what is my name your name is me which is is a slightly exaggerated example but it just really wanted to make the point like it will use its reasoning where it's capable right and and and confabulation uh is is a kind of a technical word LMS hallucinate in the sense that they can they can kind of reason things in a way that they end up coming up with false information you know um we have to be careful about that right we have to be very cautious about a lot of these characteristics of LMS they effectively I mean I would highly relate it to someone who has anterograde Amnesia right is where where you know someone cannot actually recall what has happened uh in the recent past and nothing you tell it actually gets committed to long-term memory and so I've kind of crudely used the phrase like the llms are kind of like Talking Heads uh you have to repeatedly tell

everything in every prompt and what's going on when you interact with GPT is that a lot of this conversational history is being stored in the site in some kind of cache and being resent and you're not seeing that process because they want it to be a nice user interface to where you can more easily use it so recent agentic Frameworks and workflows abound in recent Recent research yet the issues presented by limited context Windows confabulation Etc still abound lens with these conversational histories have been considered to be age in various cases yet in order to try and overcome some of the issues that I pointed out there are a variety of Frameworks that are being developed you know so one idea is an actor critic framework you could have two or more agents you give them different role prompts you tell one of them oh you're a creative you know scientist and the other one you say oh you're you know you're a harsh critic and and expert in literature on uh this scientific topic and then you have the first agent try and innovate and come up with something and then the the second agent critiques the former agent you know so you have this kind of way of like oh in order to better ground the information that one of the so-called agents is is creating you can um you can use another agent to kind of critique that make it more uh yeah grounded to reality um another uh way of doing this is building up databases of retriev documents that is you could you could it's it's an attempt to overcome the limits of context windows so what you can do is you can repeatedly query llm and any true information as well as any information you provide yourself through your own documents your own prompts uh you know any kind of formal documents you want to submit you could kind of collect these in some kind of organized database and you could query that database in a way that repeatedly send into the context window here's only the information that needs to be set right now based on everything I have and then that database kind of acts as your ground truth um that that's a means of doing it it's it's rather sophisticated there are different ways of doing that we'll get to that soon and finally there's a thinking or reflecting framework I'm just spending a second on that it's where you have an llm uh come up with things but then have it revise its previous responses U very similar to an accurate critic framework it's just you're viewing it as one one so-called agent doing this as opposed to two um in all cases the prompts are still incredibly important so what's in a prompt well uh there are many people who have now you know uh coined the phrase prompt engineering which is at this point has kind of become a field or even art in and of itself this is a recent recent systematic survey prompt engineering uh that that covers literally dozens of techniques uh for anyone who's you know interested in that um the there are many ways of doing this furthermore some other problems include uh difficulties handling numerical or symbolic

problems so intuitively you know if you read 2 plus 2 is 4 is a string as opposed to being able to recognize oh these are two numerical values uh that need to be added be an addition rule to compute a further value you know it might not be tokenized properly um and so that the LM doesn't actually know how to deal with that despite the fact that to you it's very obviously just a mathematical statement um so llms are getting better at dealing with math much better actually but they still risk this kind of in Precision so if we want to actually incorporate LMS into software applications especially those that have Stakes right those that have real world consequences where you're actually employing them in any kind of like in depth industry related setting in relation to what John was saying in any kind of ecological setting uh you know situation where you'd want to really carefully take of things verify the information and so on you need everything to be much more precise and trackable than uh whenever you whenever you uh use llms in your application you could have your application such as you know if you're developing this in python or Julia or otherwise you could have that uh code actually handle the math which is something we typically already do um as well as any kind of like if you're able to hard code any kind of rules that you can extract out of the text and actually consider those to be kind of you know programmable and then you can have them in in your program and and have that logic be carried out outside of the llm and you can just have the llm handle the natural language aspects um so kind of my proposal and this I don't want to spend too much time on this because we'll see some diagrams that actually represent the ideas here but uh you know I argue that that we should be thinking about things like pre-structured crops with what I'll call empty spaces uh for anyone who uses python is very simple you just create a bunch of preconstructed prompts that are structured in a way that you can basically Supply uh The Prompt here you're you don't have to do this word for word but you're effectively saying here is what I need here is how I need it how I need your output to be structured here are my current you know thoughts or beliefs of what's going on give me what I need uh but then make it an F string to where you can have kind of these empty spaces in between the prompt uh to where at any given moment in time say in an agent-based model you can send the most recent updated information through the FST string literal so it's like you're you have a really nice prompt that really gets at what you want the LM to send back to you but if you want to have this run algorithmically uh you know in a way that you let it run autonomously it needs to be that the information in that prompt needs to be repeatedly updated again like like reminding the the person with anterograde amnesia what's going on now then an hour later what's going on then so on um bringing it back to active emergence agents we

know they change their minds through perception they act to change the world they actually have capacity for learning including online learning learning in real time you know updating their internal model parameters might be likened into synaptic learning or modulation in the brain uh you know updating their beliefs about the relationships between observations and states the efficacy of their own actions you know a lot of these other kind of cognitive mechanisms that llms lack so uh to bring it back llms if they were like agents I would say that they're like agents who cannot act nor learn they only observe and infer uh there's been one argument that they do act it's just they they don't quite U you know that because they don't learn anything it doesn't quite matter and they can't really observe that outcome of their their action it's like their action is they output you know words and sentences and and that doesn't actually come back and impact them necessarily in any kind of tangible long-term way um they are typically trained once very expensive process some people have tried to specialize LMS by only training them predominantly on medical texts to make a kind of a medical llm or legal texts for legal llm you know but still we're stuck in this like lack of online learning or or something like clear preferences from a neurobiological perspective LMS have very broad yet immutable long-term memory and uh you know they're trained on a lot of information and then they have a highly variable short-term memory it's their context window which if anything happens to that and it's gone then they've lost you know all context um and so I argue that we should think of uh llms uh you know as as a language processing tool I think they're very effective at that they're they they absolutely outperform us in terms of being able to feed them many many texts and then you know quizzing them on those texts just in a plain as day they far out you know they exceed us and this this part of why they're so popular now uh and being considered for software applications then from this vantage point we can envision augmenting agents who learn and act such as active inference agents with something like the capacity for natural language is how I'm I'm currently phrasing it so this is kind of my diagram uh diagrammatic you know attempt at formalizing this for for anyone um these are a lot of the design principles that are going into our biofirm that we're developing right um I started developing this whenever I considered how could I have an active inference agent write a scientific paper by prompting an llm like on its own right right so so each action of the agent was you know uh prompt the llm to find more literature and citations uh another discrete action is uh you know write or revise your abstract you know and these other aspects of writing a paper then seeing if it can kind of you know choose for itself what to write rewrite update so on and how would that work and so a lot of that thinking has kind of gone

into this project um in this case what you would have is you know for for a user of the biofirm they would be able to kind of uh you know submit their own documents PDF text this is kind of the grounding truth representative of you know the users own preferences their goals what it is they want to accomplish that gets input into structured prompts you know FST string literals like the the empty spaces that way it'll it'll basically take your inputted information and restructure it in a way that's needed to get the kind of answer uh and the kind of response that you need right so the documents are sent to structured prompt uh which is then itself sent to the llm which processes it and returns your structured response structured responses uh you know allows encoding for agent observations meaning you could in a very simplistic example you could ask an LM is this a good idea yes or no and that can be encoded as a zero or a one and then the zero or one can then be you know read up as an observation for a particular observation modality that that an agent a PDP is equipped with right that that's a this very kind of Elementary approach uh but it's I hope it maybe explains the idea for for programmers who who are interested in these kinds of things um and so uh you know that observation is then taken in and and then the agent can infer States per usual What It Takes the lm's you know feedback into account it's decision making and then we have expected free energy meaning you know um the free energy it computes condition on its preferences right so it's like saying oh the agent can take in an lm's you know natural language based feedback uh to the users initial preferences as elicited and language and the agent can use that feedback to make decisions in order to follow through on its own preferences or goals right so you have this kind of reciprocal relationship between um the the user and via the llm by cariously to the the agent and then back again as you can uh take the agent's current internal beliefs you know it elicited this policy it it now believes this it previously observed this those can be put back into a structured prompt uh which could then you be sent to an llm for a new structured response and that might you know that might be something that involves like reporting like the agent is you know in natural language reporting its beliefs about what's going on and then that can be useful to the user and then be sent to uh what we'll for now call A A Knowledge Graph apologies they just now realize that this is being covered um yeah and so then this knowledge graph and we'll get more into that I was very happy to hear some other people making references to to knowledge graphs uh during the Symposium um you know it can be done in different ways but a general sense of a knowledge graph that this is kind of like your database of of knowledge that you're gathering over time in terms of you know natural language based knowledge so it's kind of like you're storing the context of the moment you're kind of storing

the truth you're storing the history of what's going on in terms of these agent llm interactions uh you know and then that way the user has the means of reviewing for themself in everyday terms what has played out during the course of simulation so that's kind of the design principle and then the knowledge graph this is a way of kind of overcoming The Limited context window issue so in a general sense uh and I'll I'll mention I'm you know a lot of my thinking about the knowledge graph comes out of this really like very recent exciting work uh being done by Marcus J buer uh who works at MIT right now um you know he's been trying to find ways of using llms to to you know um basically go over many texts in order to find patterns in the text that can then be you know kind of put together almost in this like a sharian like Innovation Way of like uh just recombining materials that already exist in text but maybe humans themselves have not yet made those connection themselves of oh uh we know all of these many different things uh if we look at it this way that actually leads to a new innovation that we could come up with you know uh we we just didn't make that connection before but we did now it's like oh you can have llms who are highly you know sophisticated whenever it comes to to going over large amounts of text having them make those kinds of discoveries um so anyway that that's the kind of where this is coming out of um and then uh sources for updating the knowledge in the graph may include user documents upto-date environment or resource reports agent beliefs domain knowledge regarding to your use regarding your use case llm responses to structured queries we kind of looked at that before um business model as a shared narrative it doesn't have to be a business model but you know it could be you have many in the context of the bio firm which we'll get into shortly um you know the idea is agents are working for a firm and so they have this kind of they need to have this shared narrative between them that kind of captures all the crucial core elements uh describing what are the operations of the firm what are the guidelines what are things that they need to attend to uh you know regulatory or legal standards um what are what what are the their current financial statements what resources do they currently have available uh what are what are the perspectives of all their different stakeholders you know being able to synthesize that information into you know kind of a singular document that's that's short enough that it but but nonetheless uh Rich enough that can be repeatedly sent back to the llm to remind the llm this is what we care about these are our constraints these are our Central concerns right now or otherwise so anyway uh I'm getting a little ahead of myself but it just that's how we can start to develop a Knowledge Graph overcoming The Limited context window by having a graphical structure to the knowledge you can keep it nicely organized and uh as we know with

graph structures with having nodes and edges you can have along the the the edges can kind of be the descriptors that relate you know multiple Concepts as nodes right so you can kind of Traverse this knowledge graph to make connections between all the entities within it uh these are the visuals here actually Daniel created these based on some particular LM uh structured prompts that that he ran and then kind of analyzed um and so the these kind of principles are going into it so by using pre-structured prompts and querying our knowledge graph um you know we can repeatedly kind of communicate what is the situational context of what's going on uh in the moment as well as here's the knowledge that we've accumulated so far that can help with problem solving during the moment this allows us to filter down only to the needed information and synthesize it reducing the final amount of information uh in this case tokens which also if you're using the open AI API to use the most uh recent like state-of-the-art models where we're usually spending at least a few pennies per llm uh uh prompt or well it depends but the the point is by reducing the amount of information flow needed you can keep costs uh significantly lower as well and then also um kind of uh quickly overcome that issue potentially exceeding the the limit of the context window and I you know I'm I'm I'm interested in finding different ways of describing this I'm a particularly a large fan of Yori basaki a neuroscientist NYU who does a lot of work on on neurobiology and and has even worked with friston at a point in the past but uh uh so I I included him to try and give a metaphor of this is this conceptually can be likened to theories of episodic memory in the brain uh for example this this uh one Theory uh that the hippocampus in the brain is something like a pointer so whenever you try to recall you know an episode in in your life you know that involved uh you know certain many episodes like in order to tell a story of what happened yesterday starting from the morning and then what did you do uh you know at noon and what did you do in the afternoon so on like the hippocampus is a kind of pointer that that kind of kind of or almost like a librarian that goes through your long-term memory to to kind of pull out sequentially what's needed in order for you to tell that story or narrative in the moment um or from a kind of more industrial or a modern data analytics example you could think of this as querying a database SQL or mongod DB meaning could be a structured or unstructured database where a report is synthesized via joining inquiring only the necessary tables or catalogs of blogged information needed for the report so again it's practical structure that allows us to kind of more quickly get the information we need without having to pull too much more than what we need um so now this is example of I'm I'm just currently phrasing this is like agent foraging and knowledge updating so it could be uh starting from the top left

we have our homeostatic agents uh John made a reference to them earlier um the sense is that these are agents who have their own preferences for you know maintaining their own homeostasis they might be trying to maintain the homeostasis of their environment in the case of an agent who designed to take care of an environment which is what we'll see in the biofirm uh they'll have their own current beliefs they're being updated over time uh you know we saw kind of an image of the the inner dynamics of what that looks like for a PDP earlier they might have their own goals which again in this case would probably align with the idea of maintaining homeostasis you know based on all of those factors uh that might lead to you know some pre-structured prompt leading to something more specific so in this case this is an agent who um wants to take care of an environment uh and this is an agent who specifically is looking at how to take care of the soil on a a natural Farm uh like a land preserver or Farm uh and so they need more information about the soil structure so they query they prompt the llm water causal determinants of soil structure and Farms you know this like an initial this is this is whenever you're first making the knowledge graph you would want to have this uh this knowledge at the very beginning of a simulation right so that it's kind of building out your knowledge graph to start so you could ask it this you could tell it to site your sources or you could integrate some kind of API uh such as semantic scholar or other apis try and check if it's actually pulling from real citations again we have that risk of confabulation making up seemingly true information uh and and so we want to uh just kind of keep a check on that so the llm tell you a lot of natural language based information about soil structure the determinant different kinds of manage pra management practices that could be used to take care of it uh different kinds of conditions and relationships between the soil structure and other aspects of the soil what's around it all these different things and then from there we could kind of go back to the knowledge graph store this deconstruct this information you know IDE ideally this would be we'd ask the llm to structure its output in a particular way and then we could send those different aspects of its response into different areas of our knowledge graph um so soil structure determinance management conditions and then also track the citations the knowledge graph again it's almost like a threedimensional structure in that you could add the time Dimension that is you could time stamp when all this information is collected uh as well you know so it depends on how sophisticated you want to make your your knowledge graph this is the last diagram I have related to this concept but I just want to include it this is what I mean by um a structured prompt with empty spaces right so we have that same agent from before um now we're sending a structured prompt I believe this so This Is pre-structured Right these

these these curly brackets kind of denote like these are one of the so-called empty spaces I believe the whatever your current environment variables are are currently here's what the agents beliefs are but our goal is here's our current business model and our current constraints are here are current constraints what Solutions exist to address this this is incredibly short and simplistic just for the sake of illustration but the the idea is you could input you could you could keep these environment variables stored in you know your local environment you know python is storing and processing them um you know the the agent beliefs are being extracted from the agent you could re-encode those with natural language you know you could you could include information like oh this agent's belief is representative of the quality of its environment so then this gets read as a sentence that's something like the agent believes with a you know 0.2 probability or very low probability that the current state of its environment is good you know you could you could re-encode that so this is really a conceptual kind of move to try and find ways Bridging the Gap between qualitative and quantitative information uh at no point in my saying that all of this is the absolute best way to do this uh what I am saying is that this feels a little bit like Frontier work uh the llms are bringing up the question of how can we properly integrate integrate them in software applications and I think these are the kind next steps in terms of our thinking of of how maybe we should be approaching this so and then you can see again uh for the rest of the the diagram we still have this communication we can continue updating our knowledge graph and also pull information back out of the knowledge graph to send into the structured prompt Um this can all be sent to the llm uh you know and and so the llm could do you could you could request it to return your response in multiple ways one way is you could tell it explain to me a natural language uh you know what Solutions exist to address this and then also ask it in the same move uh you know also give me uh you know these encodings of of of your answer so so could be that that the LM response some kind of natural language explanation of you know how to how to handle this problem but additionally uh here's structured encoding you requested Denoted as follows uh a positive one approved recommendation to carry out available action six might be the case that this uh your your agent has an observation modality meaning it has like a sensory receptor uh for the llm where a one corresponds to like an observation level of like a yes or an approval or you know go for it basically and then and then six is kind of like a second observation modality where the LM tells you which one of your available actions you should take so this is just an attempt at trying to find ways to modify an agent's architecture such that it has a kind of receptivity to uh llm outputs uh in a way that could go from natural language to actual

directives over what to do given the current context and all the information you've asked of uh that you've provided to the LM uh in the structure prompt Now we move to the biofirm um so the biofirm like this is a lot of the principles I've described this far uh kind of play into this um again a lot of work has been done on this by by by Daniel fredman here in the the talk and um and then a a lot of initial kind of coding work was done by myself um and then we've of course been under Guidance with Partnership of of John clippinger as well as David Lovejoy with uh first principles first uh I really uh like a lot of kind of the media and writing and and uh podcasts that they've been doing and so um I I would strongly suggest checking them out uh and uh it's been a pleasure working with them so far and uh so with the biofirm we actually we have a significant amount of code largely built out uh you know and kind of organized by Daniel uh with the under the active inference institute's own GitHub repository currently in a sub project biofirm um we also have a whole another library that we'll look at soon uh but for now this is kind of the foundational the starting point of the biofirm tool highlights right now we're building active inference agents as a you know using pmdp which is you know essentially has been has already been referenced earlier in the Symposium a kind of Port of uh of SPM the the mat lab Library developed by you know friston and others um and so uh you know it's been a really good library to to kind of work with and quickly start trying to implement some of our ideas the open AI API um many of you who have worked with LMS and and and tried to integrate them into code like are already familiar with this um but basically it's a way of quickly being able to prompt within your your coding environment uh a wide array of llms uh including the most recent ones and you know the new chat GPT or llama or otherwise models claw it and en cursor which you know this is not necessarily essential to developing the biofirm but uh it's just uh Daniel has kind of opened my eyes so to speak to to the wonders of of kind of integrating for developers um being able to have uh an IDE where you're editing code and then you can use an llm to help you write that code and obviously you always want to go back and look and make sure that it wrote the code that you actually intended but uh it it can it can you can dramatically reduce the number of times you need to hop into say stack Overflow for example if that makes sense to to programmers so it's great for productivity uh and then also some major things Daniel put out is um these uh live streams that cover a lot of the development of the biofirm so far um both of them are huge I mean we're looking at a couple hours or so of of material and we have close direct Code walkthrough today originally ideally I might have gone through some kind of code walk through um one I don't think we're quite there yet in terms of being able to to um we can

demonstrate the principles but in order to have something that's kind of like within the context of okay we're doing you know the active Ence Symposium uh and and we kind of need to keep things streamlined we're not quite there yet but you can get a very big picture of what's going on in that code and and get an actual code walk through by having a look at these live streams that that Daniel put out so the biofirm excuse me um uh essentially in a biofirm agents are working in a firm is like setting the scene uh to maintain the homeostasis of an environment be a Farm Natural land preserve and it could be it could be many things but for now it's I'm trying to kind of constrain uh to kind of an example um agents prefer both in a colloquial sense they want this but also in a sort of basic and prior computational sense to observe themselves maintaining their environment H stasis their own Survival and health in this sense are directly linked with environmental uh survival and health mapping internal states to inferred external States meaning their own internal beliefs are you know they're trying to model their own beliefs uh directly as you know kind of these these proxies for what's going on outside of them in the environment so they're they're very directly trying to map themselves to to their environment all of that being conditioned on preferences again what do they prefer homeostasis right so they they're trying to model their environment internally for themselves while simultaneously trying to keep that environment within some kind of H homeostatic range we'll get to a more concrete example what that means um but for now we can just think of it as like um you know that an agent's own homeostasis involves realizing his preferences relatable to how humans burn act to maintain their body temperature oxygen and blood flow since it's safety security whether it be physical or or economic um so in this case minimizing their own uncertainty condition on preferences maintaining homeostasis is equivalent to M minimizing uncertainty about if the agent is successfully keeping the environment in a homeostatic State that's it's kind of a one sentence you know concise way of looking at that they're minimizing their certainty about if they're able to successfully maintain the environment of homeostatic State uh from there we uh you know ideally what we would do is move on to building a knowledge base for example a kind of Knowledge Graph uh which itself would in an initial setup phase would involve uh User submitted documents uh so for our example case we actually did submit both a real document that John supplied uh related to uh this this Farmland uh kind of natural Preserve that he himself has taken care of of course it's been redacted um but uh the point is that this gives us a kind of real world example that we are quite familiar with uh containing a document that we can use you know for an initial testing use case um and then um you can also do other things

you can send in specific kinds of like structured queries uh you know asking for more domain knowledge or information about say how do we take care of a farmer otherwise and we have all of that then we follow that up by yeah structured prompting another idea we have that I won't go too much into right now but one idea is that you could actually extract out of natural language potentially like sistic reasoning based rules uh that there's a you know researcher axle constant who's done some work uh in the legal domain and looking at uh kind of basian uh probability and graphs and relating that to syllogistic reasoning I think that would be a really nice way of concisely extracting natural language rules turning them into something kind of like symbolic you know uh objects in Python that can be used as formula for for doing different kinds of reasoning that that increases precision and reduces the need to make so many llm calls and having the LM do that but it's a thought um anyway in any case all this all the whole development of the knowledge based Knowledge Graph is motivated by agent preferences and beliefs um so social science and I don't want to get too far away from that um you know I've been talking about a lot of design principles but I think the biofirm itself as currently it is an open source uh you know sort of resour it's still heavily under development I'll get back to you know the aspects of what's going on with that a little bit later um but but currently there's a lot of De developing development being made and is being written in a way that we ideally in the future would like this to be adaptable to different circumstances it might be the case this could be quite useful actually for continuing social scientific research the form of an agent based model um you know you could consider that these might not necessarily be agents in a firm you they might not have a a a business model they might have something else they might have you know a constitution uh you know that that links all of them or otherwise um for now I'll continue working U focusing on the biofirm but I just want that to be born in mind that I don't want to present this as if it's a highly specific use case that you know doesn't go that doesn't generalize you know allow for generalizing Beyond itself um but for now we'll stick with with uh focusing on it um so social science we have agents who are collaborating in coalitions of agents and networks of Agents uh we have you know attempting to say that we can augment agents with with natural language State belief reporting uh with consequent development of a cultural uh and this is Dr clippinger view you know if we have agents who are motivated to take care of land and then they collect uh knowledge that that corresponds to that and then they're using and communicating using this knowledge you know that then this kind of becomes something like a culture so we're looking at something like a kind of cultural grammar or or set of scripts that are being developed uh by via these agents taking care of uh

you know the the the the land around them and and we have shared narratives and then on top of that we also you know in a realistic scenario we would have agents who are doing economic resource management and potentially trying to uh you know generate some kind of value in some kind of way basically replenishing or regenerating sources available to themselves or otherwise and then on value generation and this comes back to the very start where I refer to utility as being kind of like what was previously the penultimate sort of uh uh incentive for for agents in traditional agent-based modeling now if we shift that to where now we're looking at expected free energy as the thing to be minimized wise um you know and in correspondence with kind of the the principles of act difference then uh what we're doing is that we're looking at this is you know like the approximation of uncertainty condition on preferences uh agents are trying to minimize that value where there uh with value generation a whole another thing we could do is look at each of the agents marginal contributions to the minimization of this quantifiable metric uh that is uh expected free energy uh for the whole firm like look at their marginal contributions to minimizing the uncertainty of the firm and then that itself becomes kind of value you know is we can accumulate that view that is kind of like uncertainty reduction is what generates value so that crucially this accumulation it's not some arbitrary floating value who whose own value you know changes based upon kind of Market speculation Dynamics or otherwise instead the moment to moment value generation car that that that occurs during the course of a simulation of a biofirm it reflects the agent actual learning and capacity for taking care of their environment as proxied by expected free energy and then the the kind of broader multi-agent setting and Future Vision for the biofirm their cultural aspects again as I mentioned earlier this kind of an Ever evolving accumulation of shared knowledge again we could have time stamps that also means that we might need correction of that knowledge over time um we'll talk about that more shortly um uh anyway the the evolution of the knowledge graph it's motivated by mirrors agents motivated to accomplish Collective goals for homeostasis from various perspectives um you know you might have agents who are specialized in one way agents in another you might have multiple stakeholder uh perspectives that are being fed into that initial Knowledge Graph or or along the way during the course of the simulation uh ethical uh ideally would um it would exhibit these kinds of principles like by accumulating different perspectives between agents between users stakeholders uh you know in the case of Biore Regional modeling where where there can be very very many people potentially involved in in looking at uh or being affected by how an environment is uh is being is being cared for taken care of or managed or governed

you know we being able to actually submit these these potentially conflicting or at least different uh perspectives from various stakeholders those could be submitted to the program could also include Financial regulatory or sustainability related knowledge um you know that can allow for comparing and synthesizing various perspectives again in some some kind of after critic framework you know is I mentioned earlier like a lot of methods have been used to try and overcome the issues of llms and so we be remiss not to to uh employ those in various ways where appropriate that way we can better ensure that the information that's being used and processed and being fed into agent decision making and and so on uh taking care of an actual environment uh we can make sure that it's much more accurate coherent also identify faulty or contentious information if the Agents come up with a very aggressive policy you know prior to its implementation could be critiqued and revised from an ethical regulatory or legal standpoint and all this is collective uh you know we can kind of synthesize the information I already touched on this earlier resynthesizing core information or primary information in the knowledge graph into being some kind of summary business model Constitution set of protocols protocols or missions that the agents have um and then uh sustainability Resource Management um you know making sure that agents are aware of available resources uh combined with quantitative analysis you could you know if you're looking at a real world setting you'd be looking at things like price information forecasting supplies needed for a particular policy might have qualitative reasoning involved such as why like which supplies are needed for a task what purposes uh within what guidelines um that's something I've been thinking about as well for a related Institute project uh called farmworks uh we might talk about that later we'll see um and then heterogeneity agents agents might be preconfigured or learn they necessarily will learn over time assuming we we Implement learning and their architecture more than likely well it depends on how and for what purpose and then they might even you know evolve so to speak to specializing different skills knowledge uses allowing for agents to collaborate under broad shared goals while working on subtasks there's some nice um we were joined by m yesterday um during the first Workshop um so has been involved in a lot of this previous work and then um you know another reference to kind of the as well the industrial example that I gave earlier to how how we might think about some of these things um basic form of a homeostatic agent um this is uh just take a second homeostatic agent this is the general just kind of quick natural Language summary of its internal architecture it's it's incredibly simplistic this is sort of like a sort of a kernel of what is the initial formation of a homeostatic agent prior to any uh additional you

know modification or or changes in its architecture that you think might be required for the situation at hand for now let's say there's a single environmental variable uh what we did and um you I'll talk more about it later but the idea is that we have an environment that we've an artificial environment we generated we generated it based on the documents I mentioned earlier including John's Farmland document as well as particular prompts submitted to the llm to kind of compose what are the initial makings of our our knowledge graph and it came up with soil structure as being one of the essential uh variables that should be included in our included in our environment and then we have an agent here who's kind of mapping itself trying map itself to this quantitative value uh it is a question uh right now of you know I I can imagine those who do active inference like a question might be why use a PDP why would you not use you know a continuous time continuous State space model uh for for now we're going with pomdps uh in part because uh we're assuming that we'll be using agents to model highly complex environments where it's not so simple as to kind of follow uh simple up or down uh scale or say that the the the variable is necessarily being defined by you know some kind of gaussian uh distribution and being sampled from it U it might be the case that this is much more complex there might be changing conditions or otherwise and any case for initial testing environments is also a rather quick thing to two I will admit so um but for now uh the idea is that you'd have a PDP who can view the data over time at each time step step uh Daniel ran this one for uh actually I don't remember if I ran it he probably did uh over 10,000 times steps um and so at each time step uh the agent observes if the current value is above or below some homeostatic range which the agent prefers it fly in right and so it infers the H state of the variable is it actually if I observe low is it actually low uh if it's in its homeostatic range is it actually in its homeostatic range or is it act in reality is it really high or otherwise you know and uh the another question as to whether this is necessarily a partially observable environment since this this seem to be reflective of real data but again this allows for the space that there might be very complex relationships between soil structure and let's say uh um uh moisture levels in the soil or or mic Pro bioactivity within the soil or or other kinds of things that that that really matter here to where it's much more complex than just being able to say oh just because this went low that means I need to try and bring it back up as much as possible really fast right so uh there might be more nuance and having more sophisticated architectures might be able to to kind of parse through that problem including increasing uh inference Horizons or or including more other variables that it tries to map um and in actions the agent can increase maintain or decrease the variable uh right so again very simple a little more

details later on on how that can how that can work but just say straightforward it's kind of like the agent is trying to balance out this metric um given that a lot of it all of its fluctuations uh aside from the agents actions impacting it are coming from its relationship with all the other environmental VAR in its ecosystem so it's kind of like the agent is trying to map its internal understanding of what's going on with the environment's behavior and then the agent of course prefers to see this within its homeostatic range and so it will choose actions that it believes will help this variable to get back into its homeostatic range so from there we can view this like as a whole ecosystem of variables so this this kind of defines our environment that we're looking at right in agent based modeling setting we have our homeostatic agents and then we have our entire environment composed of all these environmental variables and over in a real world setting whether you're running a firm a farm or otherwise you know each of these variables could be mapped to you know an iot device uh they could be mapped to you know some kind of uh uh logged uh observations you know that your team makes as to what's going on with the farm right now um I guess I'll bring up Farm works again briefly we've been looking at a lot of farm uh data lately that's been collected by farmers and getting some inspiration on how they go about doing those things but I will say that this uh this environment that we have here is entirely artificially generated based upon inspiration from text um so if we think about this as something like agents are trying to solve a non-equilibrium steady state uh what is that mean I'll try and summarize by saying that we're trying to maintain the structure and properties of the farm what is it that makes a farm a farm well if your soil moisture Falls dramatically too low and then suddenly all the organic matter and it dies and then many of these other things start to fall apart well now we're not quite looking at at at at homeostatic uh soil anymore right so we kind of have like this this natural decay or dilapidation of of the system the ecosystem and so agents are having to repeat like continuously uh you know kind of actively maintain uh through control inputs meaning through their actions uh they're trying to keep these variables within homeostasis and these it's worth noting these these Target ranges homeostatic ranges might change themselves over time right due to uh you know we can imagine um the the target soil temperature range might change to to adjust for you know seasonal crop change um it also might also account for certain kinds of like exogenous or external shocks to the system the impact of climate change there might be a natural disaster that occurs or otherwise that itself can really change the Dynamics required to attempt to maintain the EOS system uh as it is or as it was or if it needs to kind of change in order to continue existing given uh what kind of shock occurred so to maintain these desired

homeostatic States the agent must uh continuously work against yeah these these natural tendencies towards dilapidation States I've already mentioned this many times these these states these envir uh variables may have complex interdependent relationships may call for agents with forward-looking inference Horizons um you know of course if we over maintain the soil uh you know put too many product in it or or do other things that might still be okay for the crops it might harm local biodiversity uh you know local animals accidentally cons you know consuming different kinds of uh um um you know materials that were that were used to take care of the soil and suddenly they die that harms local biodiversity and yet uh biodiversity might also include uh pests in the environment that themselves then harm the soil or crop growth so it's just um being able to kind of adjust to all of these highly complex variables and crucially in reality we might only be able to impact some variables second some variables and not others thus agents might specialize in handling one particular controllable variable and study its relationship with the other variables leading to actionable predictive planning another kind of everyday explanation of this is like um you know I I mentioned I'm a data analyst I work in education and so um you know I'm I'm frequent looking at a lot of data regarding um um students and kind of our institution and so on I'm based in Chicago uh some things that we can change our bus routes for our students to make sure that they can get to school safely uh something we cannot control is the amount of crime in Chicago which um anyone in the US knows is a that would be very difficult to do right so so so we're looking at there are all kinds of factors in our environment that we cannot control and then there's what we can and then trying to find can we control what we can control in ways that kind of can balance out with all these things that we cannot control right so that is very crucial to be able to identify and then measure analyze those kinds of relationships whenever we're trying to design any kind of concrete actionable uh uh policy right or intervention that you might want to make and then uh John referenced the commons earlier right and Elanor ostrom's work um homeostatic agents solving non-equilibrium stud States the ever ever solving the commons so so I'm I'm trying you know I I was thinking about a lot of these principles and ostram was required reading for me as someone's coming you know from a social sciences background and so I this part of why I find this project so interesting and why I find it fascinating to try to realize a lot of these things um in principles involved so we we have rules conditions adaptation I mean the agents necessarily have to adapt to the environment around them if you provide some kind of Kernel agent and and give it some space to figure out what it is it needs to do it's necessarily going to have to adapt right so so we have that attention towards the local um recognizing uncertainty

and adaptation you know agents will have to take care of homeostasis manage resources through learning uh to to change conditions and broader unobserved uncontrollable environment in the sense of um you know there are many things going on that the agents cannot directly observe um there are other things that will occur that were unexpected you know so and then also there might be relationships within the variables themselves that you know for example there might be some kind of critical threshold that's hit by one variable that then has some kind of domino or Cascade effect on the others you know so that that's why I include both the phrases endogenous or exogenous shots um you know being able to to watch out for those kinds of Dynamics then governing the commons you know that's sort of that's sort of the benefit uh of you know if it can be done right if it can be done successfully integrating llms in you know a very in a very thoroughly considered way that allows for you know accurate information uh uh uh accumulation and sharing and then using it for implementing policies um you know that that's the aspect that kind of keeps things grounded and then and then gives this kind of natural language based way of dealing with an environment you know this this kind of just repeats many things I mentioned before but we're synthesizing a lot of info from various stakeholders uh various domains you know that are needed to handle the problem repeatedly grounding agents of common goals from maintaining the environment against perturbation capture maintaining its Markov blanket and then nested collaboration potentially so just as users can practice oversight by running these models um agents themselves you know you might have agents who are at another layer so to speak who who maintain their own oversight over other agents or coalitions of agents and those uh those kind of hierarchical Agents might themselves act as something like an observer role um they might recognize a broader issue going on uh in The Firm that isn't being accounted for by specialized agents and so you can have those kinds of kind of hierarchical uh uh specialized uh structural roles um that that allows for this kind of nested uh nestedness and how the environment is maintained um you know and and then it's a very interesting idea that I was going to bring up later but in the context of like looking at Biore Regional hubs or something along that line I mean if you were actually employing something like a biofirm to a real world scenario that I don't see any reason why you couldn't have some way of interfacing between different biof firms right because if they're actually playing out in reality and building their own knowledge graphs you know say in real time and some kind of way then then you could very much have that kind of interfac and you could have you know structured prompts pulling from the knowledge graph of one biofirm you know sending that information to another

biofirm and back uh you know so so that's you know I I don't want to get ahead of ourselves but because the vast majority of the actual concrete work that's been done on the biofirm has only been over the course of frankly the past couple of weeks I mean longer than that for sure in terms of of John developing the ideas conversations we've had over the past couple months but whenever it comes to developing this stuff and actually realizing it it's it's still quite new so it's very exciting to to report biofirm for users let me just do a wrap okay um I'm gonna have yeah yeah I'll speed this up a little bit um biofirm fre users yeah uh one you know for for for for a user's perspective first you submit your documents as I've mentioned many times uh you know your own PDFs real world information uh you know if if you want to submit textbooks you know regarding uh agricultural techniques and methods for taking care of farms you know including formula that describe relationships something you could have from there the direction is to have there are two intended kind of potential paths you could take right now all the testing is being done with the first path but the ideal is to move it in the direction where you could have both available first one is uh for being able to explore counterfactual so we generate synthetic data using domain knowledge or historical data that's what we're already doing and then second direct application that would be where you're actually able to send in data and have it being streamed and have this play out real time this very much you know kind of in the future but that it's sort of on the roadmap so to speak um and then once you followed one of those paths uh you could generate an environment for your agents and then also you could construct applicable agents meaning agents to your design or ones that kind of fit the situation or be mapped particular environmental variables as we saw with the agent who takes care of the soil structure earlier using agent maker uh which is kind of a entire addition that that that d you kind of Zoom through uh being able to incorporate included there's more information on that uh the second live stream that was released um from there you would uh simulate uh your agent based model meaning there would be an initial uh setup phase that takes in your initial documents and so on constructs the agents constructs the artificial environment then runs your action perception Loop it keeps things moving data and llm responses are encoded as observations for agents agents take these observations take care of their environment so we already know homeostasis for the agent based on how it setup is kind of hardcoded into its design and Ma to the home homeostasis of its environment and finally there would be obviously we would want this analytics so having some kind of analytic Suites that allows you to derive insights from different kinds of metrics that are being recorded over time there might be different kinds of like llm summary reports and recommendations that

actually distracts the metrics and you know through a pre-structured prompt as I've said a million times during the course of this you know using prur prompt can in put a lot of these results and then and then have an llm kind of put it into a natural language like just explain for you in everyday words here's what happened here's what's going on here what the agents believe and so on uh Biore Regional modeling I wanted to look at this really quickly this is uh a lot of Daniel's work on uh I mean this is just an initial step but what's happened is that Daniel sent particular kinds of structured prompts you find more information on this in the first live stream uh but basically asked about information uh in very particular ways um to llms and and did it programmatically was able to extract incredible amount of information about these different regions different counties and uh he started with California and then moved to New England and then it's expanded from there and so we have like Maine and Massachusetts so for example uh Massachusetts uh Hampshire County like you can uh he sent in like you know generate some kind of uh Supply strategist statement um we have copies of these this is a Json form meaning it's kind of a set up in a way that can be read by the program then here's just a more readable legible you know every day for a user report it's like here is you know your supply chain strategist report on the ecosystem of that County uh references to what's going on in terms of its biodiversity how climate change is impacting it how land is used habitat fragmentation water quality so just a lot of the different things that are going on specific to each of these counties you can do that from an AR a market analyst perspective as as well try and consider different kind of opportunities there the current economic landscape what kind of regulatory environment compliance requirements are their state federal local of course these are currently very short descriptions uh ideally over time you would build out your knowledge graph your your knowledge base you'd actually you know kind of zoom into each of these send more follow-up prompts like tell me more about this you know site your site the information you're pulling from um and then you could even consolidate that so this is this kind of a Consolidated uh report you know very long for all of these different Market analyst perspective an ecological perspective supply chain strategist and just kind of finding ways to to synthesize a lot of um information in one move this was a business case and I just wanted to show quickly like here are you know ideas of you know here you know defining out potential revenue streams risk assessment mitigation strategies you know so it's very in fiveyear financial projections of course we could pull this kind of logic back into the program have local computations done that actually comes up with much more uh uh uh uh uh grounded realistic and precise information based on available resources that you have or that

that at least artificially the agents have if you're doing more of an artificial setting um so yeah it's just it's making all these linkages so um you know that that becomes sort of the basis the the starting point for for our um for the biofirm so we we then you know again we generated this artificial environment all we did was send in user documents structured prompts through an llm you a structured response um let me quickly yeah um this allowed programming our environment uh it's it's Dynamics configuration generating descriptions for each of the variables they get stored in this nice Json structured file ecosystem config you know so I mean we it you you saw previously like I was looking in a GitHub repository we like we were really building out the code in that in that base um and so anyway that sticking back to the artificial environment here this allows us to pull out yeah what are our variables or proxies for VAR Ables such indices like it can be quite difficult to measure biodiversity so we make an index for that there might be other ways of doing that there might be more specific ways of doing that that fits your own use case or otherwise we could you know Tor that um domain inspired quantitative relationships between the variables you can even you know with more manual control you can set the noise levels for each of these like if you assume that there might be some noisy process you know uh from a more mathematical or statistical perspective how what which of these are actually controllable or not for those which are controllable how much are they controllable can you dramatically change biodiversity in one move probably not um or you could and I don't want to follow that thought further biofirm development performance benchmarking uh this is another crucial part of the biofirm development being able to actually Benchmark how our active inference agents uh uh perform relative to you know some kind of benchmark Baseline so just having a more reactive or random sort of thermostat style like reactive um control so basically on the left we don't have active inference agents it's just trying Instead This is a a more basic uh uh control strategy that kind of just reacts to the data it's seeing uh and of course the data is getting very out of hand I'm not saying it's not getting out of hand with the active inference agent but much more so that's what we're seeing from initial test case and this active inference agent on the right is still that basic kernel homeostatic agent right it is keeping its Target within range you know 21% of the time relative to the random strategy which is only seven um you know the next steps would be to actually fine-tune and modify uh the agents to better match the situation so far it's it's it's very simplistic it's only looking at one variable only tracking one variable has it's currently not taking in any information from an llm it's currently not taking in any information about the other environmental variables um this was really interesting to see though that it already has this

capacity for kind of outperforming the the reactive model now we've had some instances where the active inference agent doesn't uh you know it maybe matches the reactive model so it's just you know it's prime time to to step up the agent development game you know so that this was just step one it's like can we can we build it you know we yes we can so now let's improve it um analytic Suite we can look at you know the the variables themselves their correlations with one another we track the metrics about the environment over time then compare how the environment behaved in different different simulation settings um you know all that's what we want to do with agent-based modeling in the first place typically we want to be able to run simulations look at the outcomes based on different circumstances and then we can compare you know how this went using different agents um and uh you know uh we could use llm integration to to to translate this into like a a more readable report for users um this is a whole another aspect I I really wanted to emphasize it I I've made this kind of as an additional module so it can be edited kind of after the fact like outside of the main program uh which which also means that you know you could you can modify this fit as you want but basically the idea is you try and measure the surprise or uncertainty or expected free energy or otherwise of the environment here um and and um basically look at how the actual variable diverges from where you want it to be you know the difference uh you know it's of course it's mathematical and strong from active inference and other areas basing in statistics and mechanics but um the idea is you can measure how far away it is from where you wanted it to be in terms of surprisal uh you know actually capturing kind of this more rich or nuanced information about how that variable moves versus where you want it to be um and then you can kind of flip that say oh well in a non-equilibrium steady state system where more usually than not the system is going to have a tendency towards generate uh generating more and more surprisal that is the the the complex um relationships between the variables is going to lead to things typically going off course Very quickly if you don't touch them um then in that case any move at reducing surprise meaning any move at combating that natural tendency towards kind of like chaos or entropy uh you know can be viewed as a form of generating value and so this is a kind of additional metric we could add saying that there's a kind of value added to having these agents uh minimizing the surprise meaning minimizing the free energy meaning minimizing the distance between homeostasis and where the data is now uh right so so we could see that kind of accumulation over time and again it's modularized meaning for a user they could potentially experiment as they like they could change how this value added is computed I think that'd be very interested interesting for people who are interested in

economic simulations or or doing this you know differently or want to see like how how does this work you know um then also shle value analysis we have shly values so um in the course of you know these simulations we have agents trying to solve or take care of their environment and so in the case of U shle value computation what it's doing is that we can compare how do certain coalitions of Agents run through the simulation compared to one another uh in terms of U you know these different metrics and so you can see like oh agent all five agents together uh you know did did better at maintaining their environment in homeostasis uh they they did better than only having um you know these this other particular Coalition of Agents but they actually did worse altogether versus when you only had these two agents working together right so it's it's a very kind of sophisticated way of looking at how agents are actually solving the problems and actually carrying out actions that they need to carry out um and we can do this you know for different metrics and we can view that as you know that that can be another way of saying this is how value is being generated from by these agents right so being able to make those kinds of you know those kinds of analytics then uh final yeah these are the last couple slides GNN and agent maker um this is very cold it's kind of like uh you know we did so much rapid development of the biofirm over the past couple weeks that this just kind of like hit me I was like oh I don't I didn't really think about that but Daniel has incorporated along with his co-authored work with Yak smel know in the past um they've worked on what's called General notation notation and and and it's a it's a way of kind of translating active inference uh active inference agents meaning models or architectures uh between different programming languages including the able to being able to compose them using natural language uh and then also being able to visualize them graphically like that's you know it's not quite there yet the kind of makings for um being able to have this kind of translatability like uh you know the ideal would be being able to write to an llm like hey make an agent for me that can do this and this and then by giving the LM certain kinds of contacts it can fill in the matrices and and so on as you need it of course we're still a distance from that but it's kind of building the framework towards moving towards that and that way you can make agents who are much more tailored to the situation and furthermore ideally you'd be able to uh translate them between programming languages meaning you know being able to switch your applications in Python or Julia or otherwise um being able to have that done for you and then uh Daniel has basically forked uh the PDP repository and then built agent maker on top of it so right now it's kind of tailored to PDP um but it could be used in other ways as well and so we already have some next steps on the kinds of Agents we want to make for me it's agents who should be able

to observe the whole environment but only be able to control their own respective part to me that's what would make the idea of like having a shared common goal amongst heterogeneous specialized agents like that's what would real that so that's kind of the broader view of the biofirm you start with inputting your documents your texts your PDFs uh data for fitting not act um excuse me that's optional right now uh it would be more ideal for a real world setting but for now just generating an artificial environment just the text is fine we've been able to accomplish that so far and then from there you could fit agents to the situation or at least maybe customize them yourself using agent maker you have the the start of the your knowledge graph or base being developed and you have that uh ecosystem all of its Dynamics and configuration kind of Spun up then after that you just run the simulation uh looking at you know analytics uh we want to move towards having an interface in future having dashboards we've already seen a lot of these um graphs earlier so it's you know and then we'd also want to have llms that could actually for the results and this is basically towards an adaptable kernel of a biofirm this is kind of the bigger picture we have different kinds of design considerations uh couple highlights we what we're looking at things like implementing potentially amortized inference we're kind of accelerating the inference process um using continuous State space predictions and models you know and then considering you know we we have an RX and fur group at The Institute so we'd like to potentially move towards moving using a more reactive framework using RX INF fur the Julia Library as well as kind of complimentary uh library for environments um the ARs environments Library uh plenty of different ways you might be able to use different kinds of apis or Integrations with other tools um locally storing LMS to mitigate the issue of having to necessarily like uh use the open API uh open AI LM API um so and then obviously integrating you know key aspects are like the resource management right and that that's that's actually I would argue is going to be one of the easier parts of incorporating because it's like just as we can currently read real data or gener generate artificial data and have the agents manage that like soil structure we could do that similarly for different kinds of like U you know Financial resource uh values or managing an account something along those lines right so bringing it back biofirm development testing is just a slide I showed earlier but it's it's just a you know reinforce here some of the tools we use some things we want to do in the future some related Institute projects related to you know we're working on projects that relate to mitigating future existential risk we've applied to a grant with Forest site there's the fli grant project farmworks ARX and fur group there's a a new economic simulations Related Group some people are interested in looking at things like you

know simulating currencies and and so on um in in agent based modeling then those live stream updates um my computers having uh repeatedly deciding to cover things St and that's it so uh we we went much further on time intended there was a lot to there was a lot to cover um really appreciate every everyone's attention though um and you know in these 11 minutes we have left you know if you guys want to give anything a shot please go for it voice is going out no take take some questions yeah anyone in the live chat can write a question John where where does this take you Andor what will you plan to do tomorrow well I think what we tried to lay the foundation here in a very thorough way is oh what is a biofirm and uh and what how is able to address uh many of the issues that you see and in people trying to work out a bio uh bio Financial to the Biore Regional economics and create the proper incentives and so um I think what we're trying to do here is really show that there's a number I started to say here are some of the criteria that we're looking for in particular building with eler aam and the whole idea of how to govern a commment is a profound issue that seem to be intractable to a lot of people um I think this in my my view this provides a framework for doing a scalable framework for doing it a way involving people a way of applying scientific methods and creating finan cial incentives and automating a lot of the the data collection analysis task and how you can agents can build up their own competencies of over the period of time so there's a lot here there's a lot to digest um and it does recommend it does provide a new metric of value that's based upon and and a you're not you're not establishing value the classic economics that our price Arbitrage or information asymmetry Arbitrage you're actually saying there's such a thing as a free energy principle a scientific process and we can show how the value is created and then how to allocate it to those people or those agents that are creating it so this is really sort of foundational so we want to take it in next steps um and what we're going to be doing tomorrow is is um actually there's a group of people who are experts in the sense that they work with uh how to take multi-agent architectures and apply them to in some cases the to the Enterprise from Del that's doing that another group uh Simon uh uh Torrance is doing it in terms of AI looking at framework for agenda architectures uh there are others are looking at it applying to the bio uh Finance bio Regional Finance issues um so we have that Susanna Choy we also have uh uh we have Matt who's head of sustainable finance and metabolic and then there's John Havens who's been involved with i e establishing ethical Frameworks for that application of geospatial models in AI so it's quite a combination of people but what we want them to to look at is what's the impact of this when you're having fully autonomous biomedic based agents uh and what happens when you can apply them not

just to the biometric to the Biore Regional Finance but also the new notion of a firm um and I think there's is a real profound issue here what happens when you really have autonomous agents and you create institutions we create institutions that are supposed to be above people but the problem is people populate institutions and they come captured but what happens if we start to put these agents and build institutions that are that are function this way that are verifiable this way and what does that happen to what we think about employment jobs and I mean there's some really substantive issues here um and how do you create this as a a public resource a public good that doesn't get captured by particularly small interests so it's a bondad spectrum of questions we're putting out there um that I don't think that the public in general fully appreciates the the the capacity and the power of an active INF free energy Principle as applied to economic and and and societal issues so that's we're going to try to tackle those issues and get people talking about it and this is this is a beginning but we want to take it next steps and really get rep reference get actual reference Technologies out there that we can actually show and demonstrate and may be able to replicate that's that's the the goal yeah and very helpful I gotta say you know Daniel kudos to you from all along I mean both you guys and Daniel just you've done remarkable work and and and accelerating this process and we're very very grateful for that and what active infes institute there's other things the other sessions I saw here that seem to be very relevant to what we're doing so I'm really looking forward to collaborating I'm I'm quite fascinated with with thought Forge with Matt Brown is doing and doing you homeostatic networks and it goes back to Ashby and the law of re a variety I think all that's applicable to this too so there's this wonderful synthesis that you've enabled you you really have I and and through a singular effort so again extremely grateful for this thank you it's been awesome and and uh lot more I can say but what what what can we're gonna we're gonna ask more of you you're I'm not done yet you're not done no no we we're no the the part that resonated with me in the presentation over the last two days of biofirm and how Andrew laid out all these pieces pieces is like on one hand we have that researcher lens again about similarities and differences and about treating it like it's an engineering problem and then I think John you've brought in the deeper history of Commons and cybernetics and these more open-ended problems about how do we connect any technical capacity to the embodiment and the co- enactment of systems that are livable and regenerative and that's a little bit of a different kind of topic that's one though that that uh mass convening and awareness can [Music] can fertilize and make real in a way where the technical understanding it's like you can read it for 20 years it's like it is just going to be the textbook it's just going to be H

something to jump off of and take into another way it's not like some of these questions about how different regions will work it's not it's not in the 2022 textbook so that's applying active inference is taking the high road and the low road and variational free energy expected free energy like all of those formalisms and then like surprise surprise applying them to what matters and figuring out that project and operational part and there's just like there's so many pieces was just like thinking of the yarn connecting the the the needs to what we do and don't have at different levels of Readiness that takes us back to the tech tree and all these other ways that we've discussed coordinating around that balanced way to think about Regional and what is Federated what is transferable how do we deal with all of these aspects and it's um been a great journey over the last few months also well thank you I I what I what I find so reassuring is that and and you you see applying the method and then it starts to explain and you keep applying it and so it it it doesn't it doesn't Fade Out it you're not you're not get o h it doesn't become ad Haw and so I feel like we're doing something really principle that is extensible and when I come from the the the bof finance community and see all the things that they've been trying to do are in the Commons Community say actually it there's a wonderful fit here and getting these two worlds together is is is is really important it's a challenge too um but i i i people want to do that and and they're looking for something like this and so I really hope that we could going to be able to create that marriage as it were to add one comment on that I think when you opened today and said well we could make things fungible we can connect but that is also part of replicating if not elaborating the problem which is the fungibility of drinking water here and this caler here and this unique species here this facilitating that fungibility is facilitating the capture of the underlying resource so even if it were somehow trivial to make like assertions about impact monetized and the whole hypers certs and impact certification even if that were trivial and we can just assume that that we have full observability or something like that it could even accelerate Financial capital based Cascades and so what what can be an escape grounding to that and that's where seeing like whoa the models on through the Paradigm of incentives are maximalist and reward oriented can they be homeostatic or allostatic oriented to satisfies and be on paths that bounce surprise rather than try to maximize but then as soon as we look around the world there's things that don't make sense we're not trying to maximize or minimize like the reservoir but rather within a range and so there's so many things where we can just connect first principles of intelligence and research questions to just like the calm subtle sometimes ways that that the technology because to to to move too fast also would would again replicate fallacies right okay well good luck with the session

tomorrow okay and and I I hope many people come join and and engage in these Fun open source meaningful projects this is just the beginning it's just a checkpoint and and uh it's a strong one and and I'm very pleased with it yeah cool all right thank you fellas see you later guys for